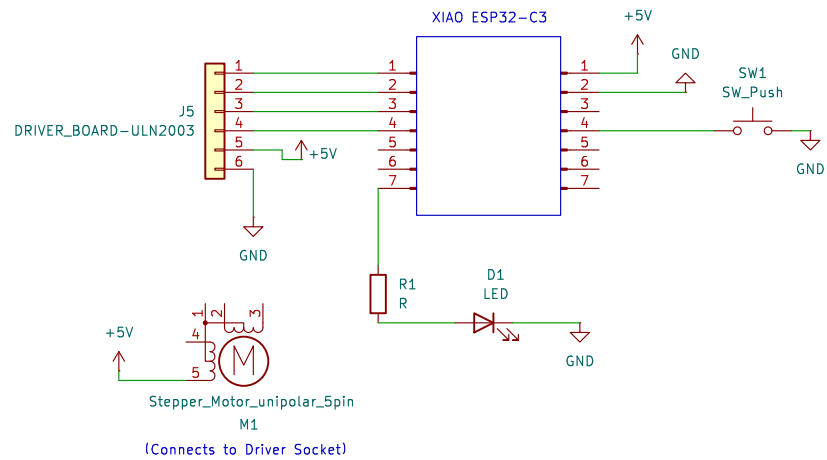
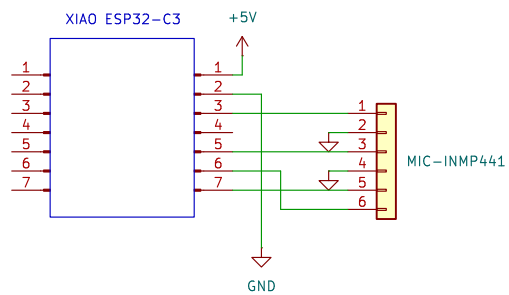




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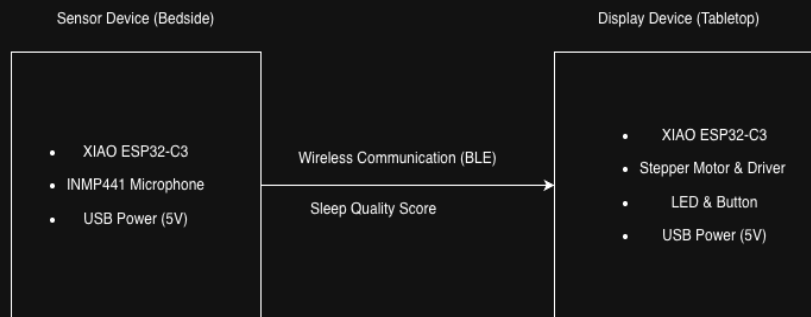
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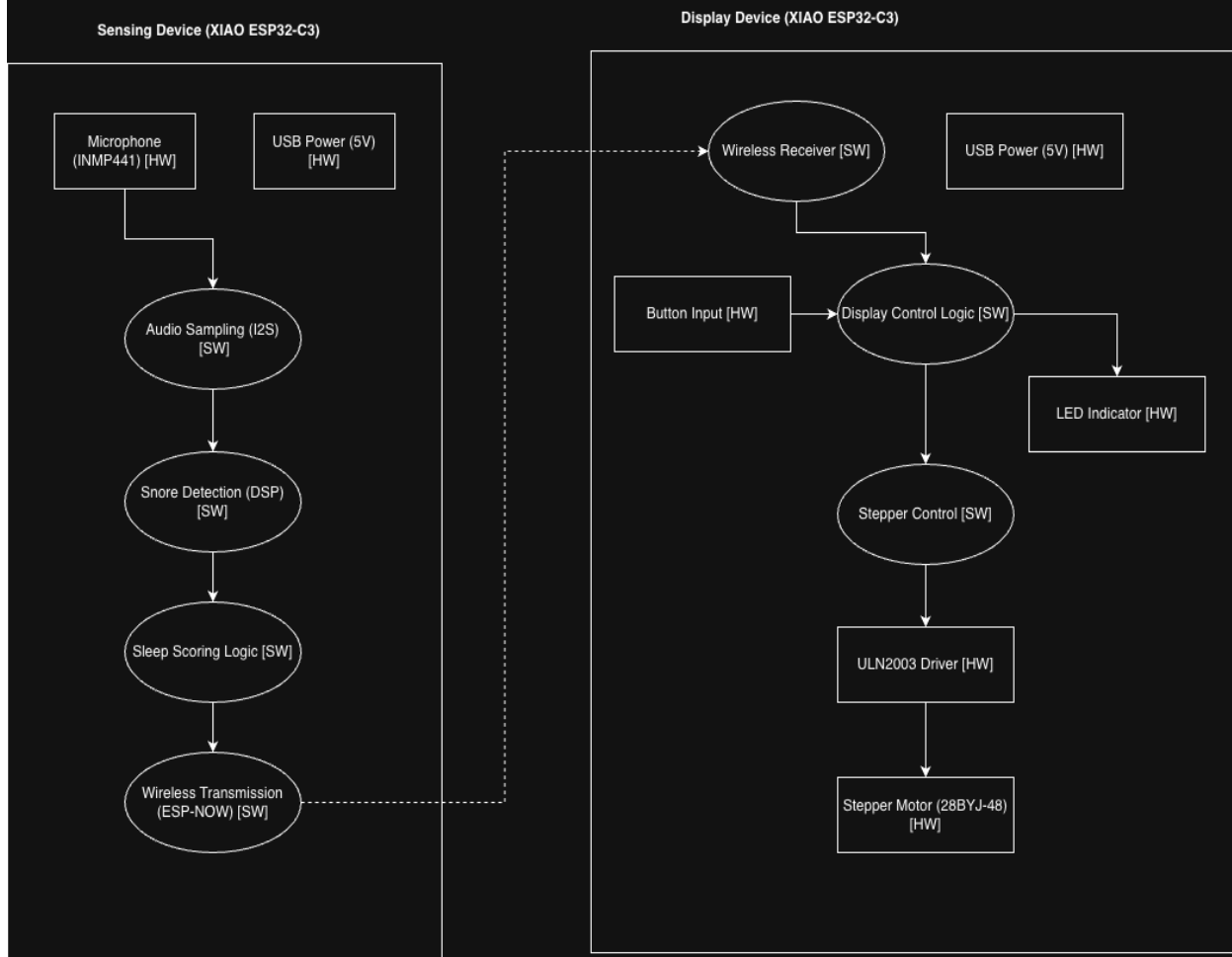
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Communication Overview



HW + SW System Architecture



Item ID	Component Name	Description	Part Number	Quantity	Supplier / Notes
1	Microcontroller	MCU with WiFi/BLE, RISC-V core	Seeed Studio XIAO ESP32-C3	2	1 for Sensor Device, 1 for Display Device
2	Microphone	Omnidirectional I2S Digital MEMS Microphone	INMP441	1	For Sensor Device (High precision, Low noise)
3	Stepper Motor	5V DC Geared Stepper Motor	28BYJ-48 (5V Version) (TBD)	1	For Display Device Gauge
4	Motor Driver	Darlington Transistor Array Driver Board	ULN2003 Module (TBD)	1	Usually sold as a kit with 28BYJ-48
5	LED	5mm Red LED (Through-hole)	Generic	1	For Snore Activity Indicator
6	Resistor	220Ω or 330Ω, 1/4W, 5%	Generic	1	Current limiting for LED (Essential)
7	Push Button	6x6x5mm Tactile Switch	Generic (B3F series compatible)	1	For Reset / User Interaction
8	USB Cable	USB Type-C Cable	Generic	2	Power supply & Data upload for both units
9	Prototyping Board	PCB	Generic	2	For circuit assembly
10	Jumper Wires	M-M, M-F, F-F DuPont Wires	Generic	1 Pack	For connections

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